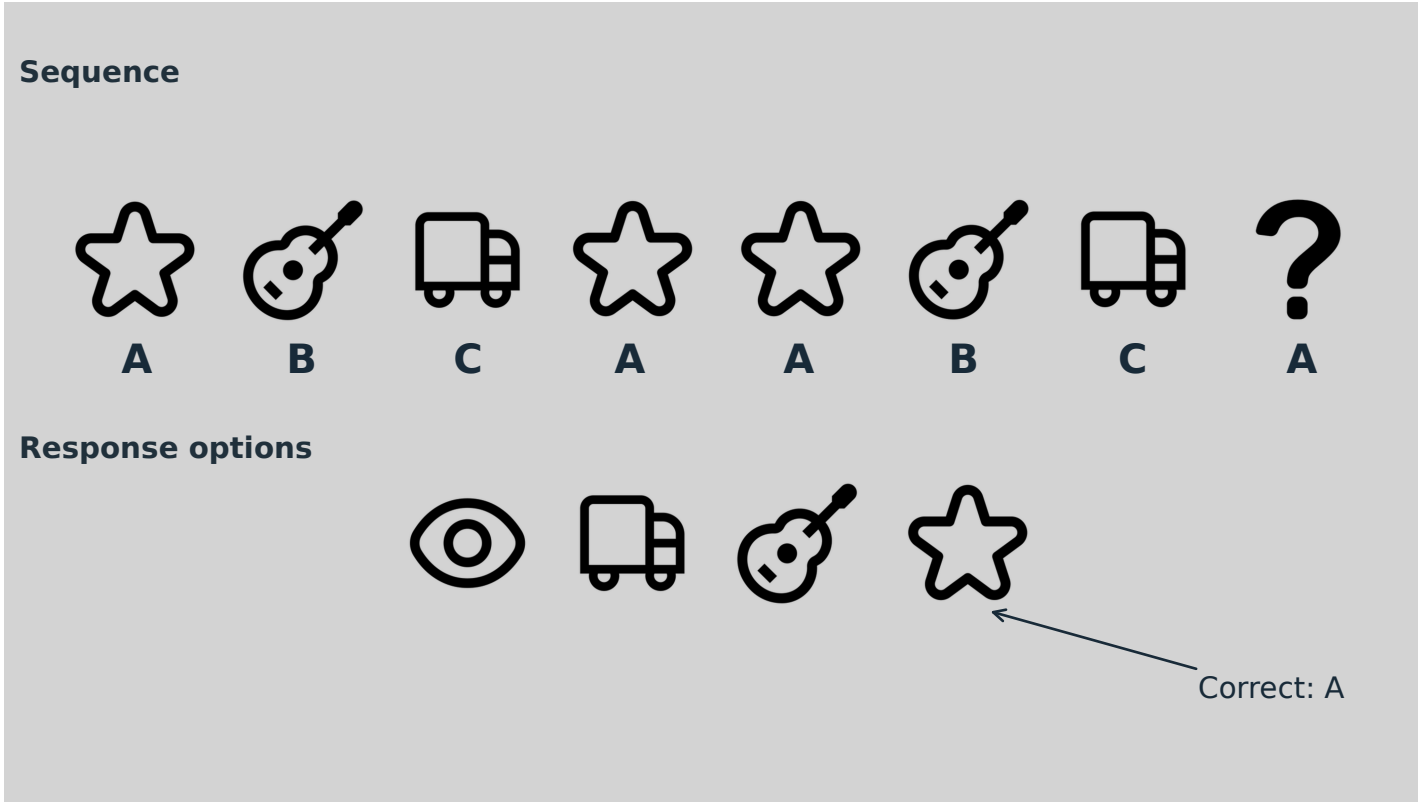
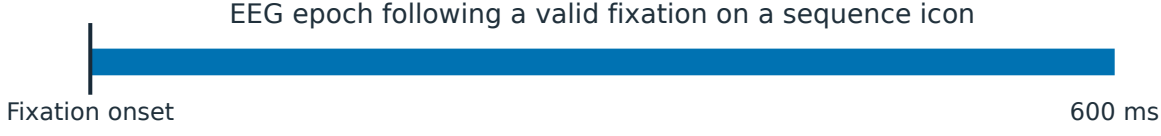
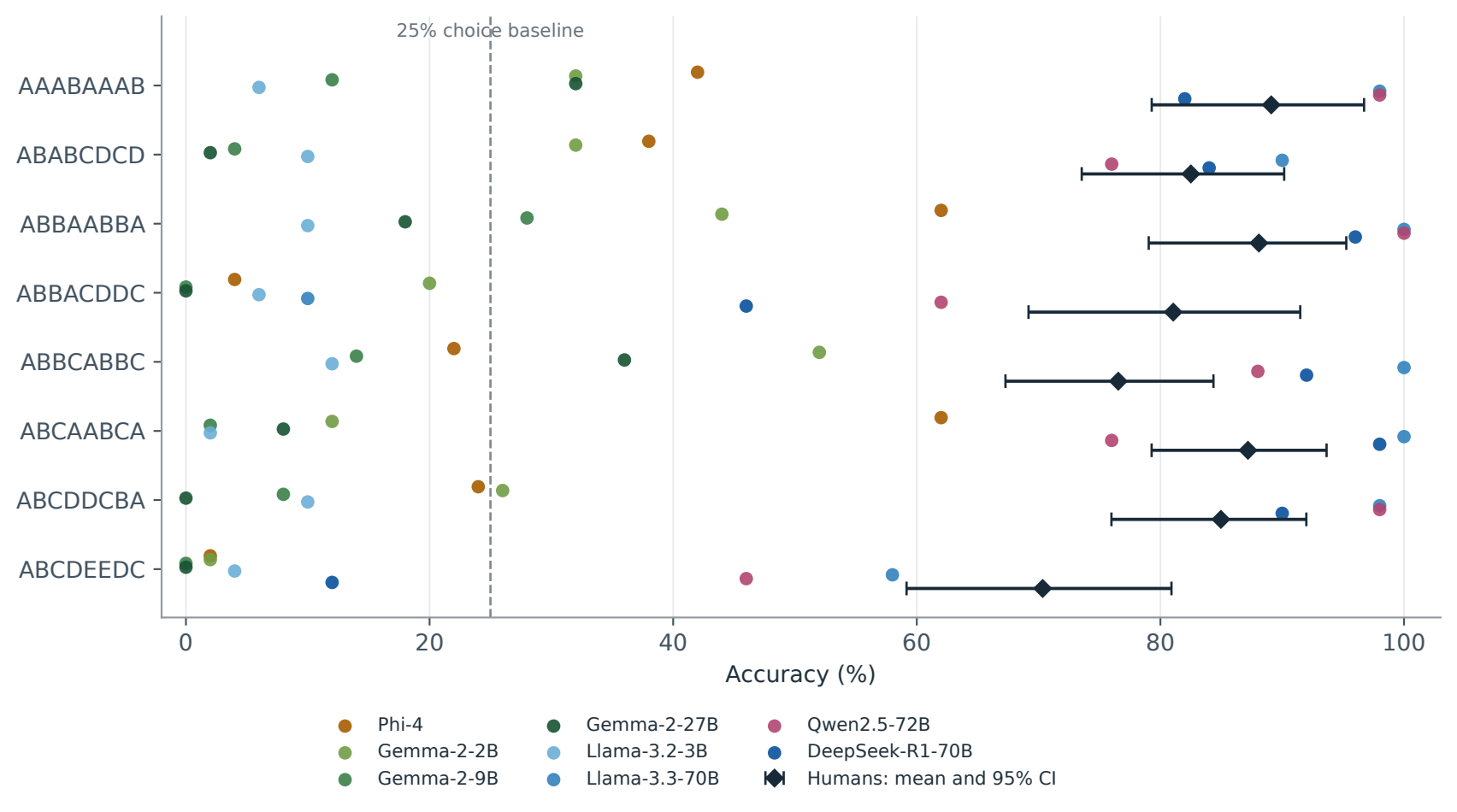
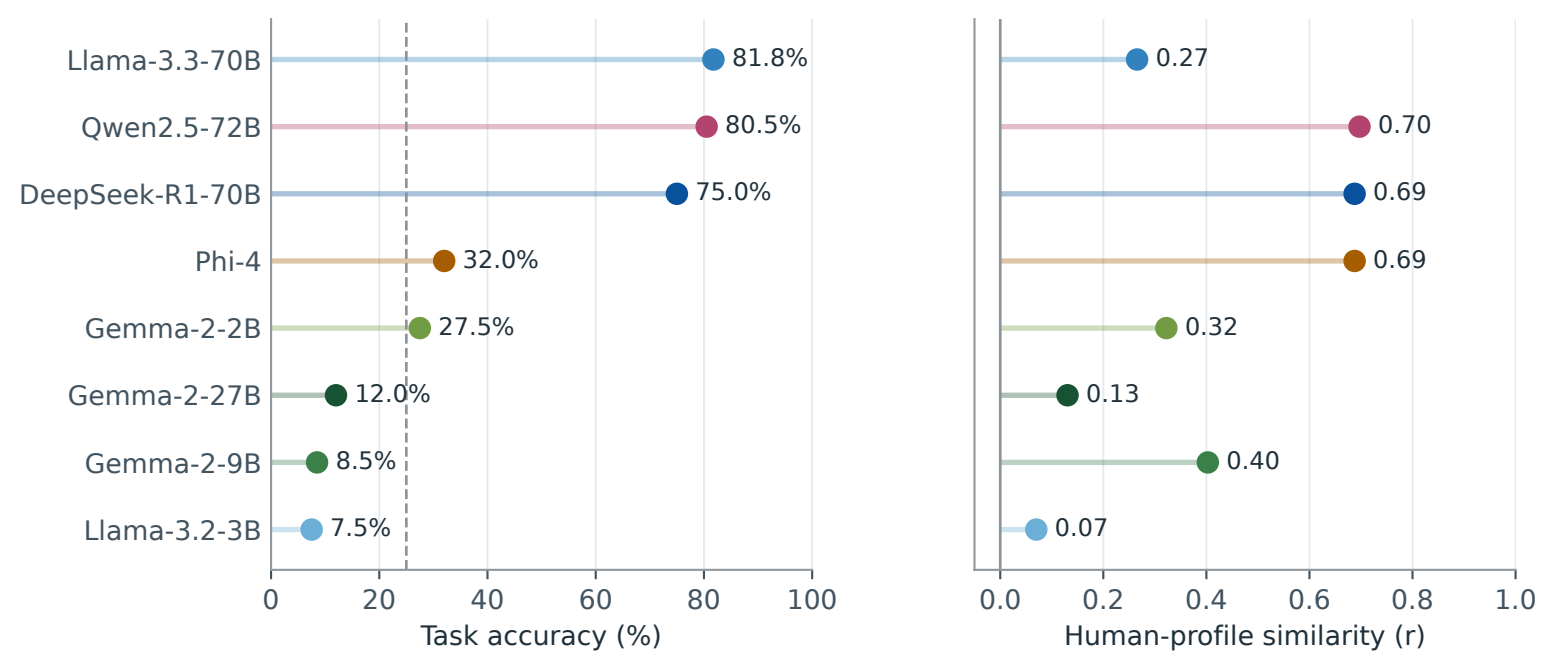
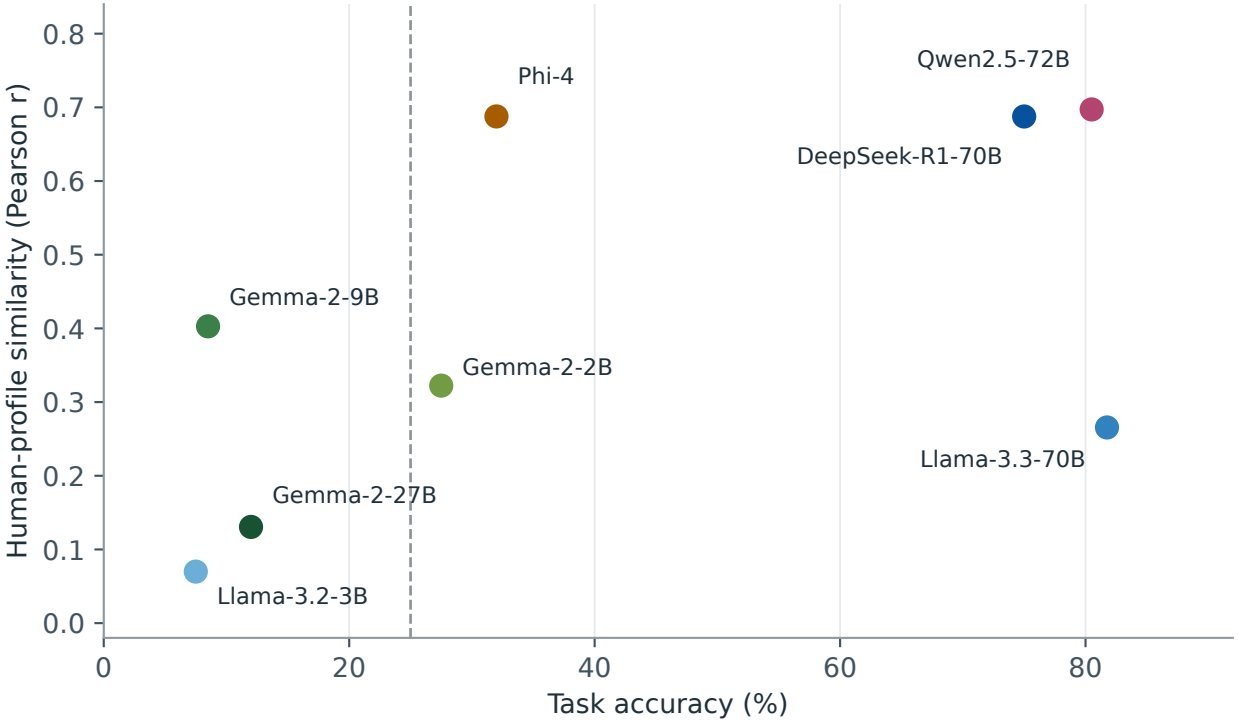
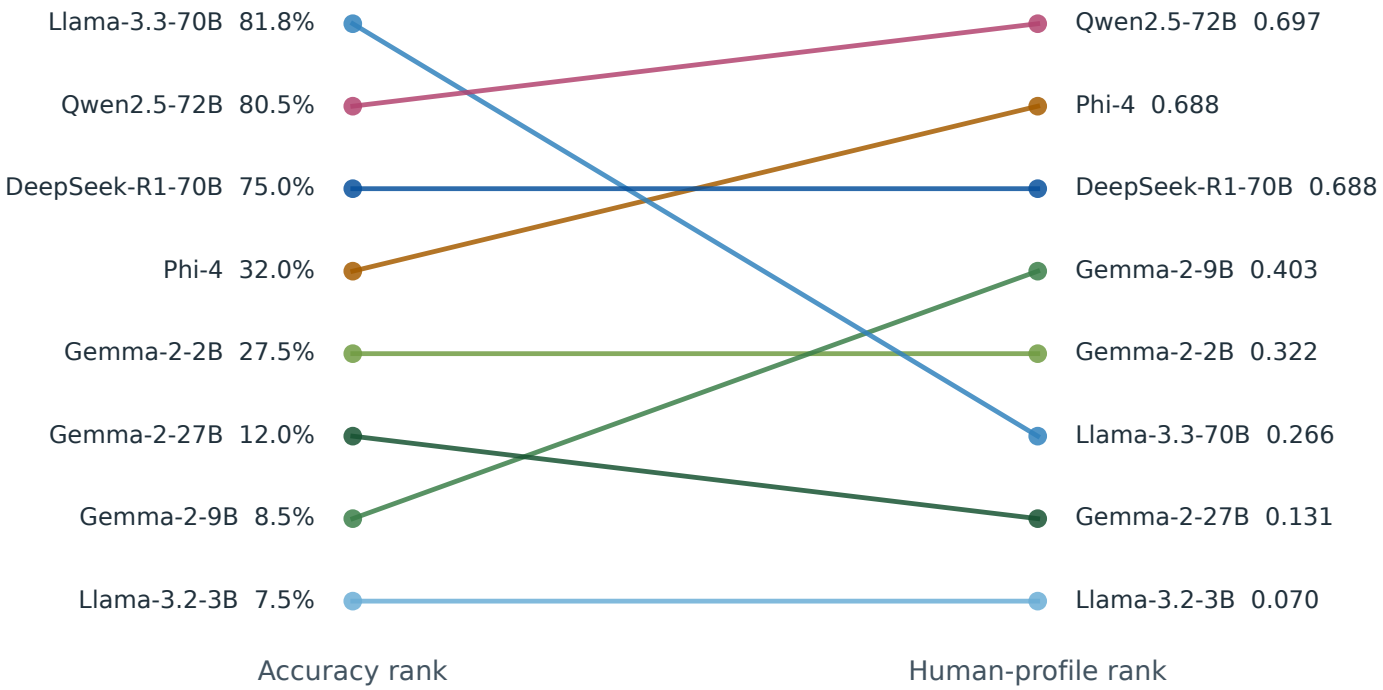

Sequence
A
B
C
A
A
B
C
A
Response options
Correct: A

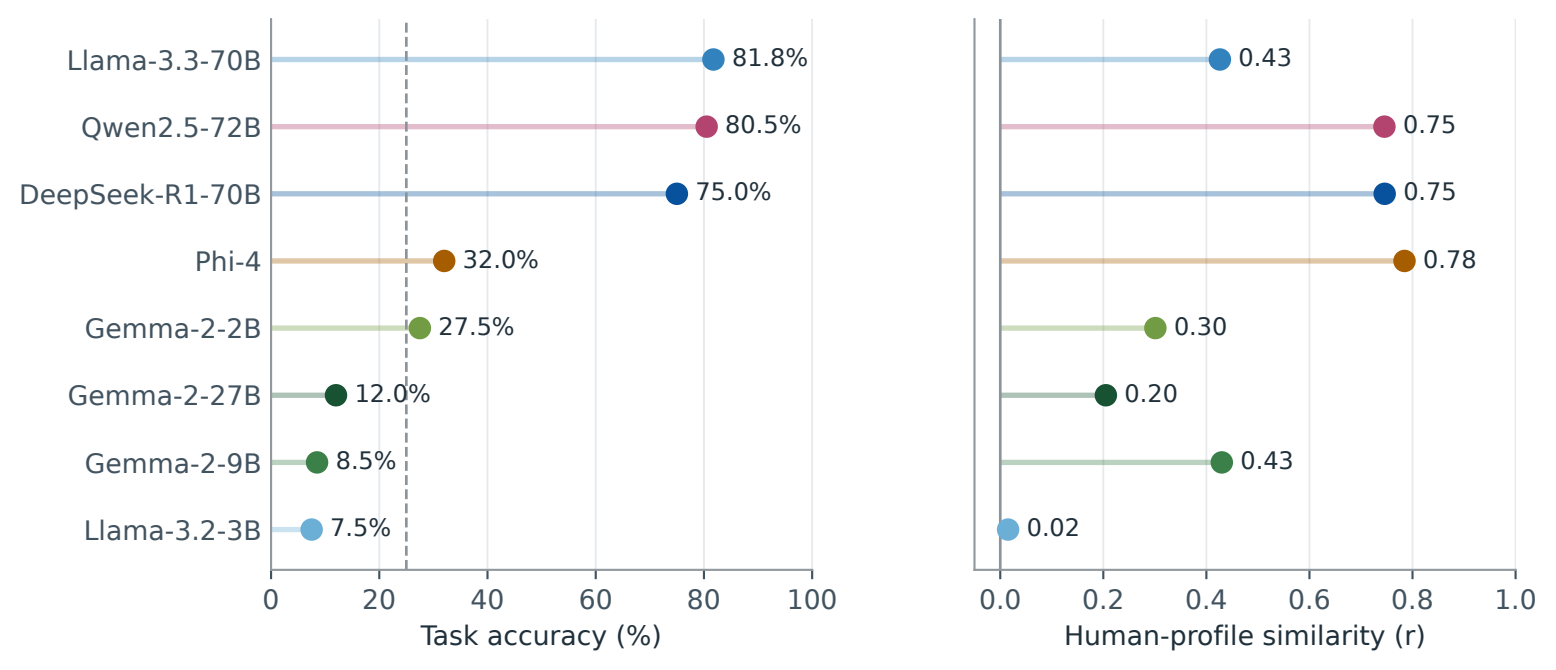

EEG epoch following a valid fixation on a sequence icon
Fixation onset
600 ms

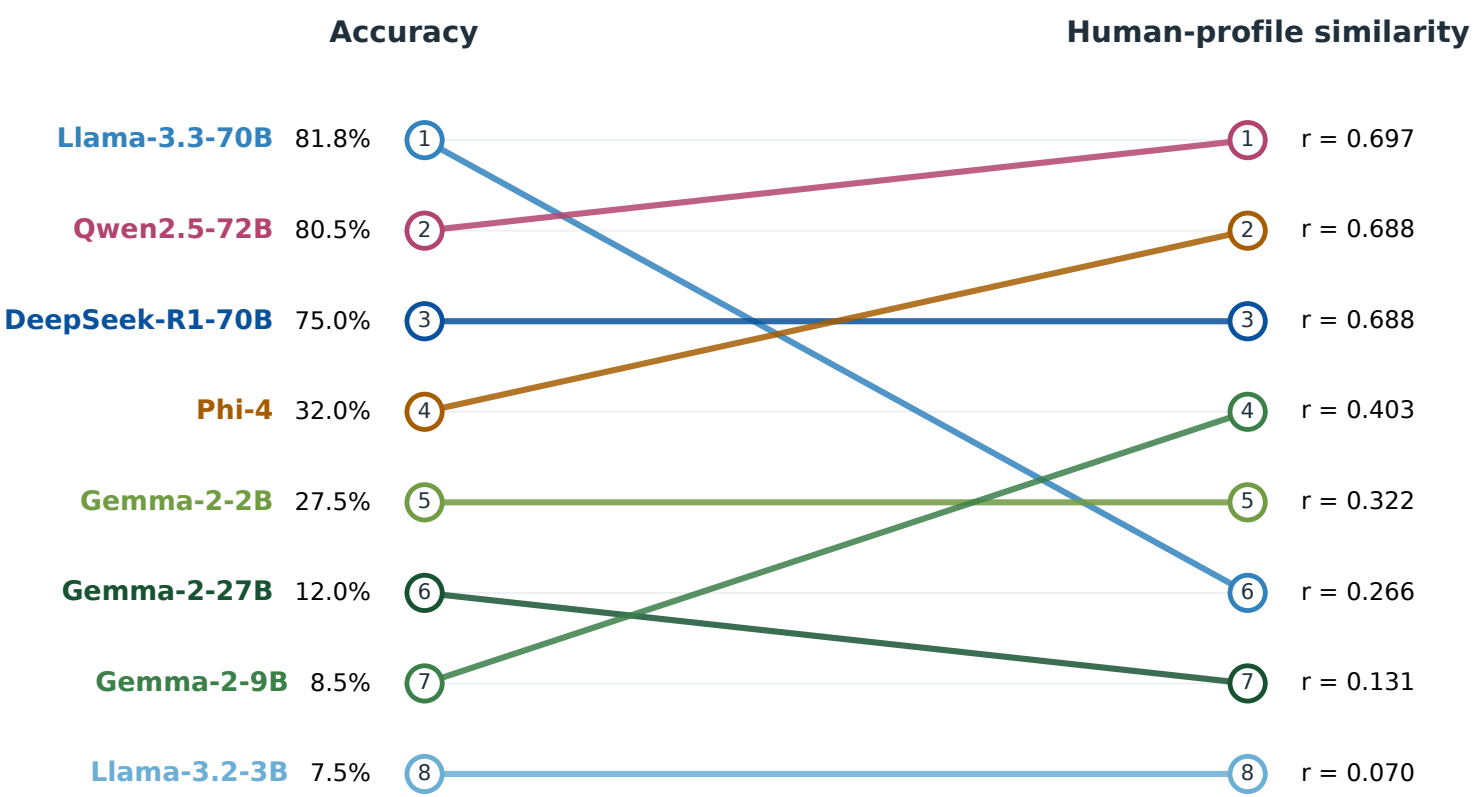

25% choice baseline
AAABAAAB
ABABCDCD
ABBAABBA
ABBACDDC
ABBCABBC
ABCAABCA
ABCDDCBA
ABCDEEDC
0
20
40
60
80
100
Accuracy (%)
Phi-4
Gemma-2-2B
Gemma-2-9B
Gemma-2-27B
Llama-3.2-3B
Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Humans: mean and 95% CI

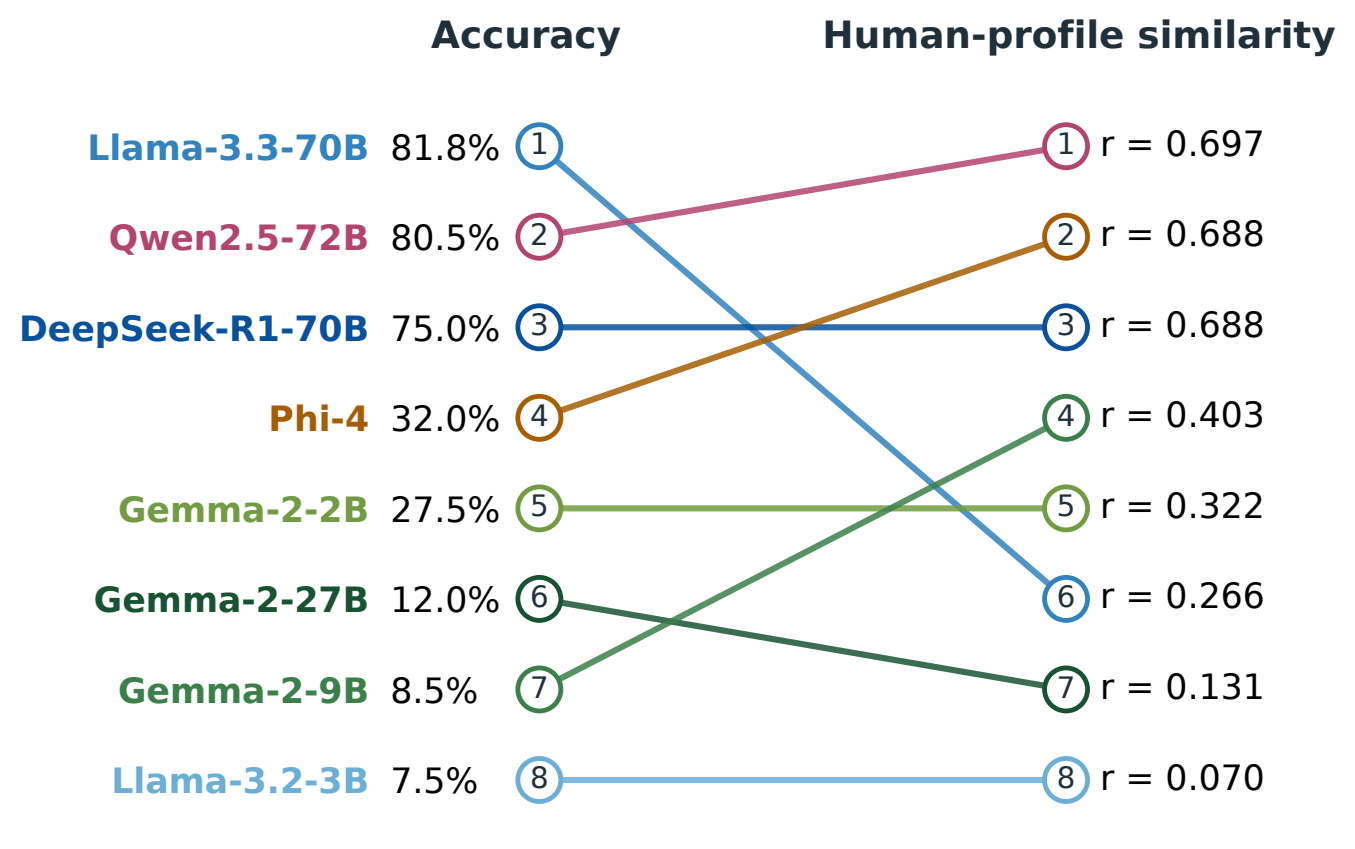

Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Phi-4
Gemma-2-2B
Gemma-2-27B
Gemma-2-9B
Llama-3.2-3B
81.8%
80.5%
75.0%
32.0%
27.5%
12.0%
8.5%
7.5%
0
20
40
60
80
100
Task accuracy (%)
0.27
0.70
0.69
0.69
0.32
0.13
0.40
0.07
0.0
0.2
0.4
0.6
0.8
1.0
Human-profile similarity (r)

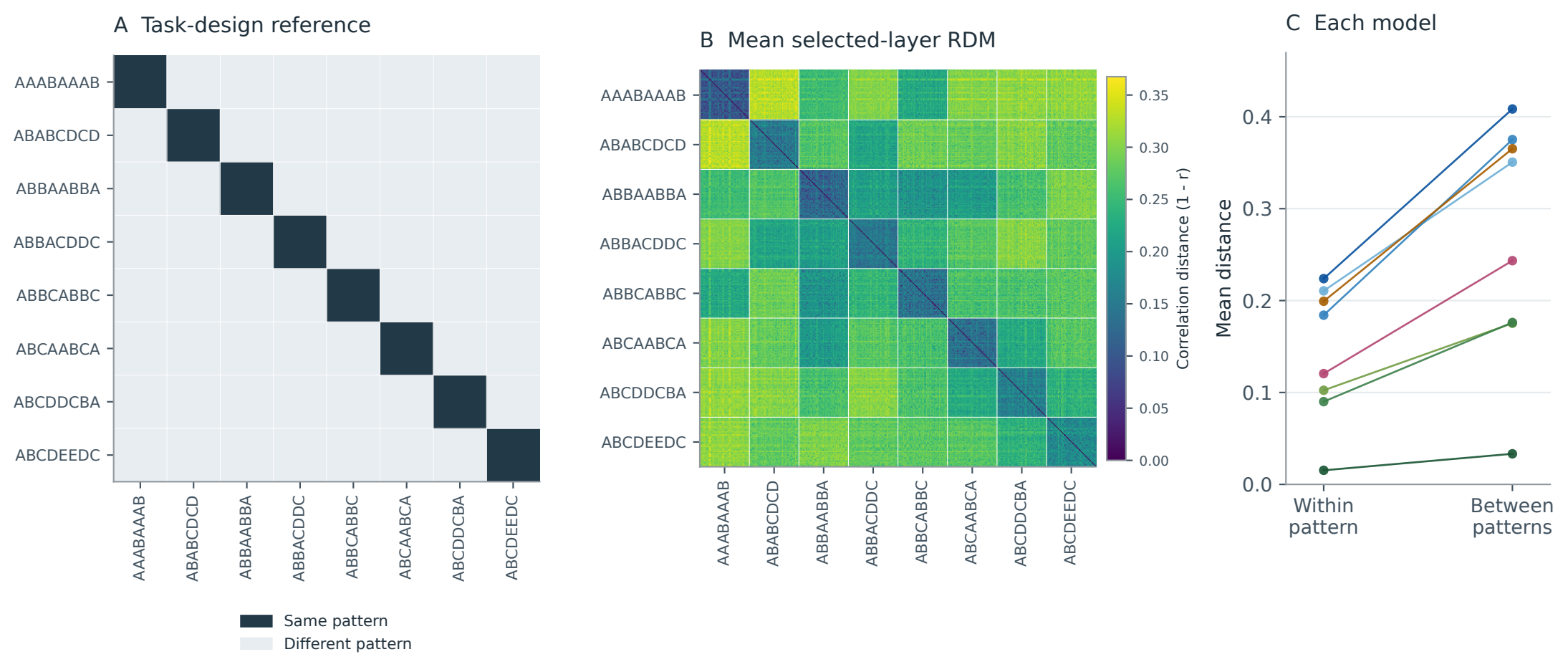

Human-profile similarity (Pearson r)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
20
40
60
80
Task accuracy (%)
Phi-4
Qwen2.5-72B
DeepSeek-R1-70B
Gemma-2-9B
Gemma-2-2B
Llama-3.3-70B
Gemma-2-27B
Llama-3.2-3B

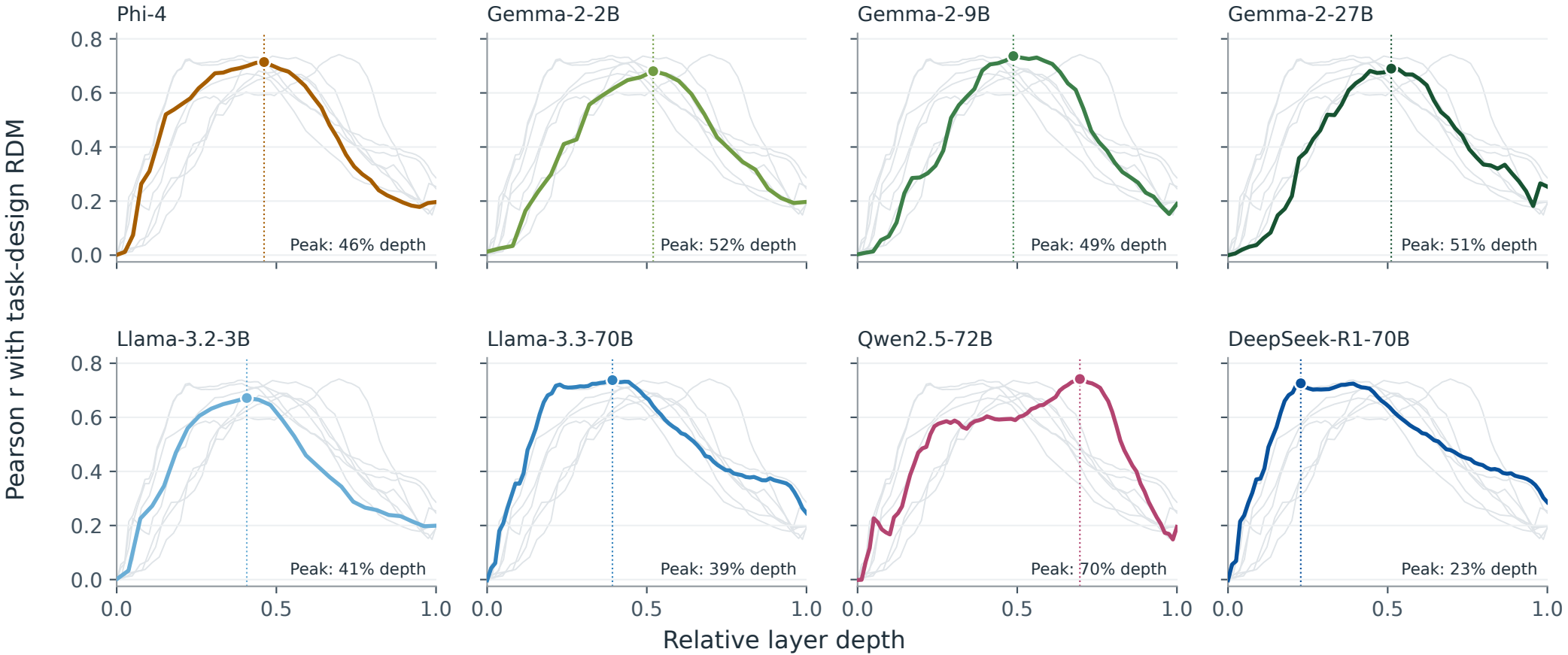

Llama-3.3-70B 81.8%
Qwen2.5-72B 80.5%
DeepSeek-R1-70B 75.0%
Phi-4 32.0%
Gemma-2-2B 27.5%
Gemma-2-27B 12.0%
Gemma-2-9B 8.5%
Llama-3.2-3B 7.5%
Qwen2.5-72B 0.697
Phi-4 0.688
DeepSeek-R1-70B 0.688
Gemma-2-9B 0.403
Gemma-2-2B 0.322
Llama-3.3-70B 0.266
Gemma-2-27B 0.131
Llama-3.2-3B 0.070
Accuracy rank
Human-profile rank

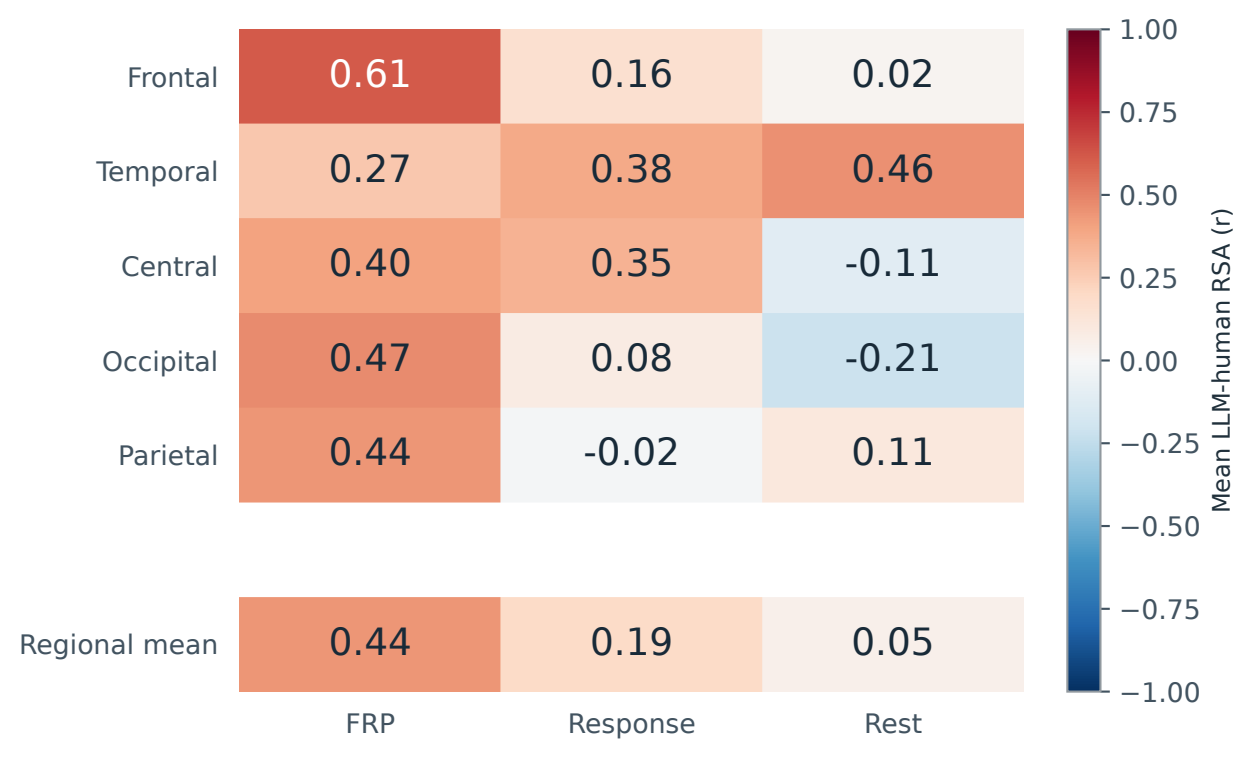

Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Phi-4
Gemma-2-2B
Gemma-2-27B
Gemma-2-9B
Llama-3.2-3B
81.8%
80.5%
75.0%
32.0%
27.5%
12.0%
8.5%
7.5%
0
20
40
60
80
100
Task accuracy (%)
0.43
0.75
0.75
0.78
0.30
0.20
0.43
0.02
0.0
0.2
0.4
0.6
0.8
1.0
Human-profile similarity (r)

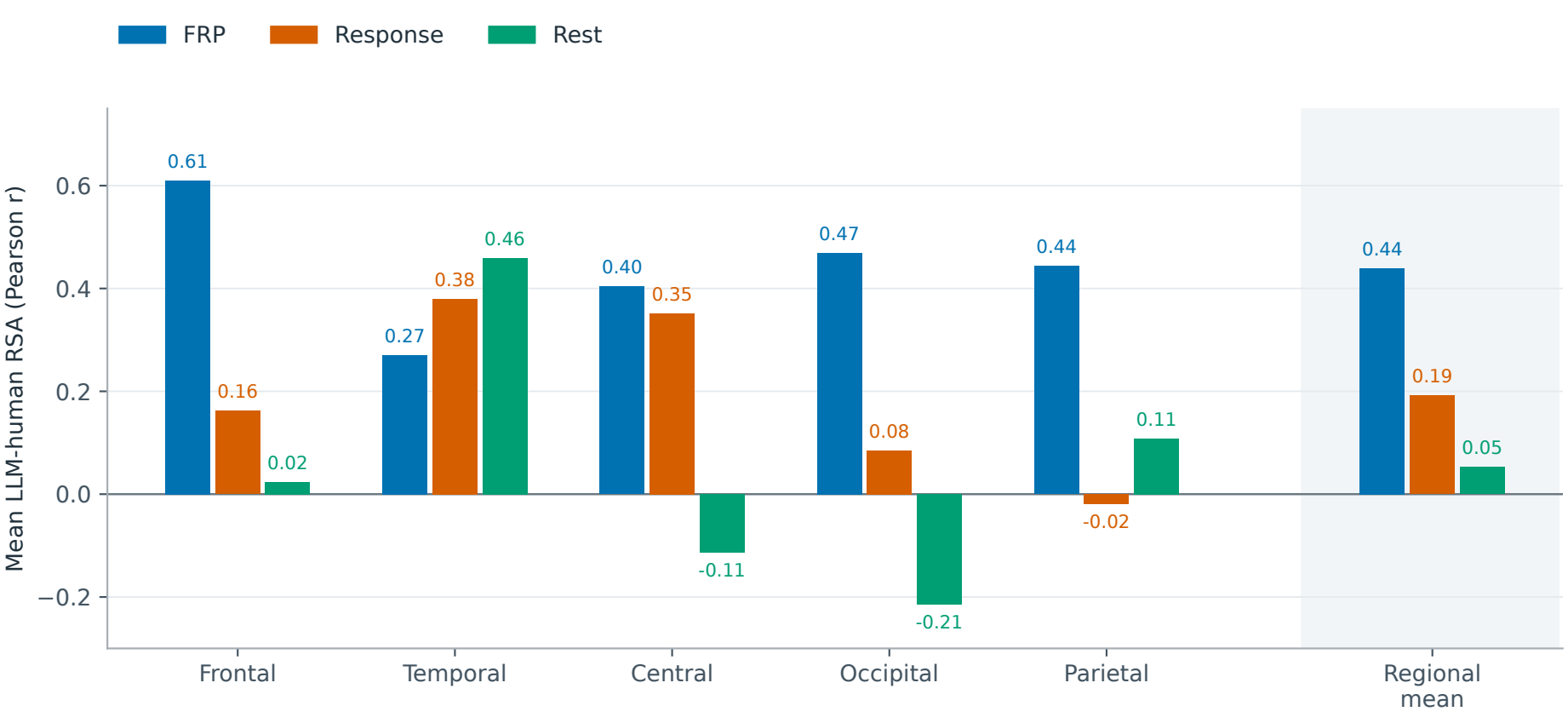

Accuracy
Human-profile similarity
Llama-3.3-70B 81.8%
Qwen2.5-72B 80.5%
DeepSeek-R1-70B 75.0%
Phi-4 32.0%
Gemma-2-2B 27.5%
Gemma-2-27B 12.0%
Gemma-2-9B 8.5%
Llama-3.2-3B 7.5%
1
2
3
4
5
6
7
8
r = 0.697
r = 0.688
r = 0.688
r = 0.403
r = 0.322
r = 0.266
r = 0.131
r = 0.070

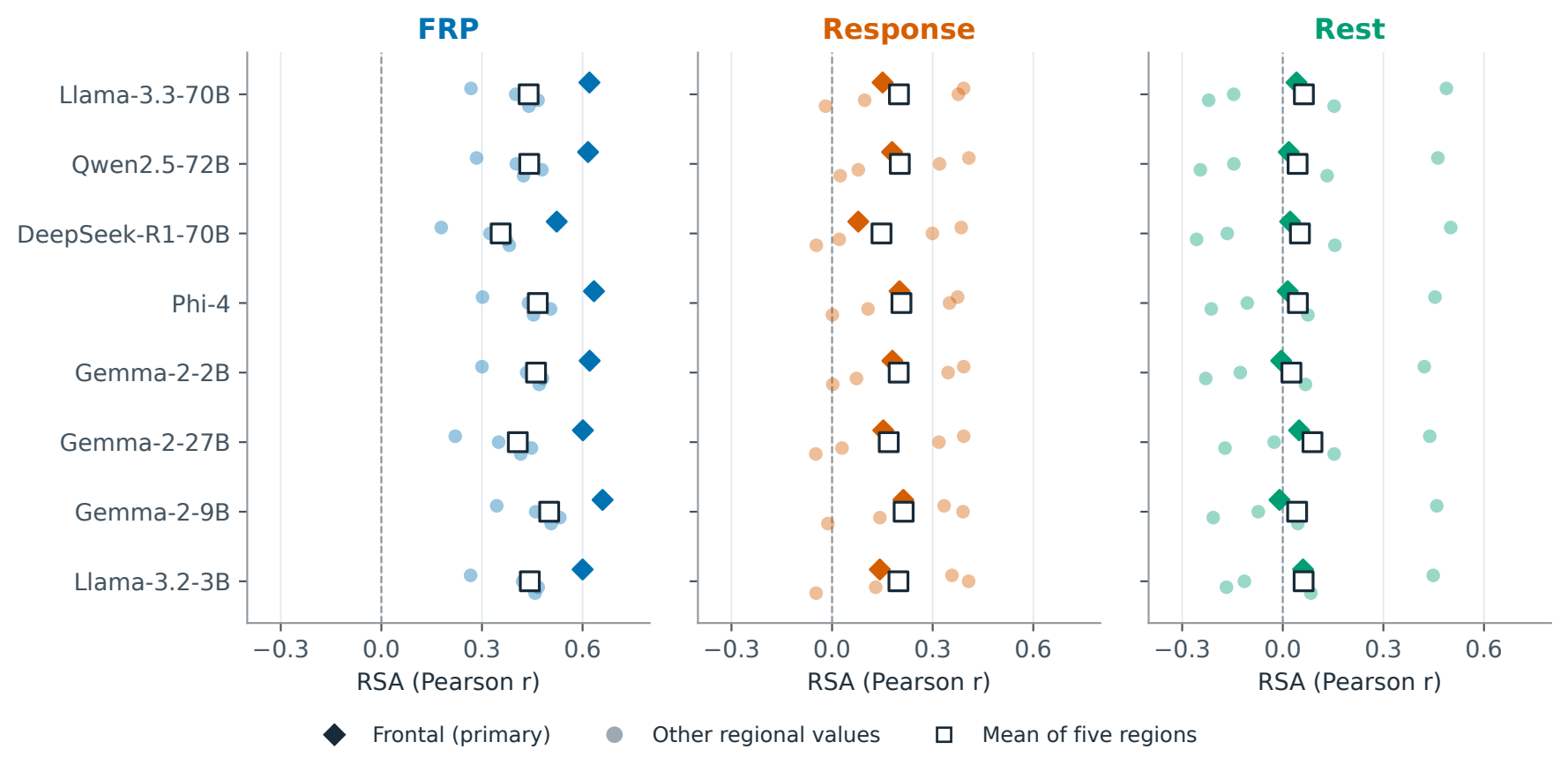

Accuracy
Human-profile similarity
Llama-3.3-70B 81.8%
Qwen2.5-72B 80.5%
DeepSeek-R1-70B 75.0%
Phi-4 32.0%
Gemma-2-2B 27.5%
Gemma-2-27B 12.0%
Gemma-2-9B 8.5%
Llama-3.2-3B 7.5%
1 r = 0.697
2 r = 0.688
3 r = 0.688
4 r = 0.403
5 r = 0.322
6 r = 0.266
7 r = 0.131
8 r = 0.070

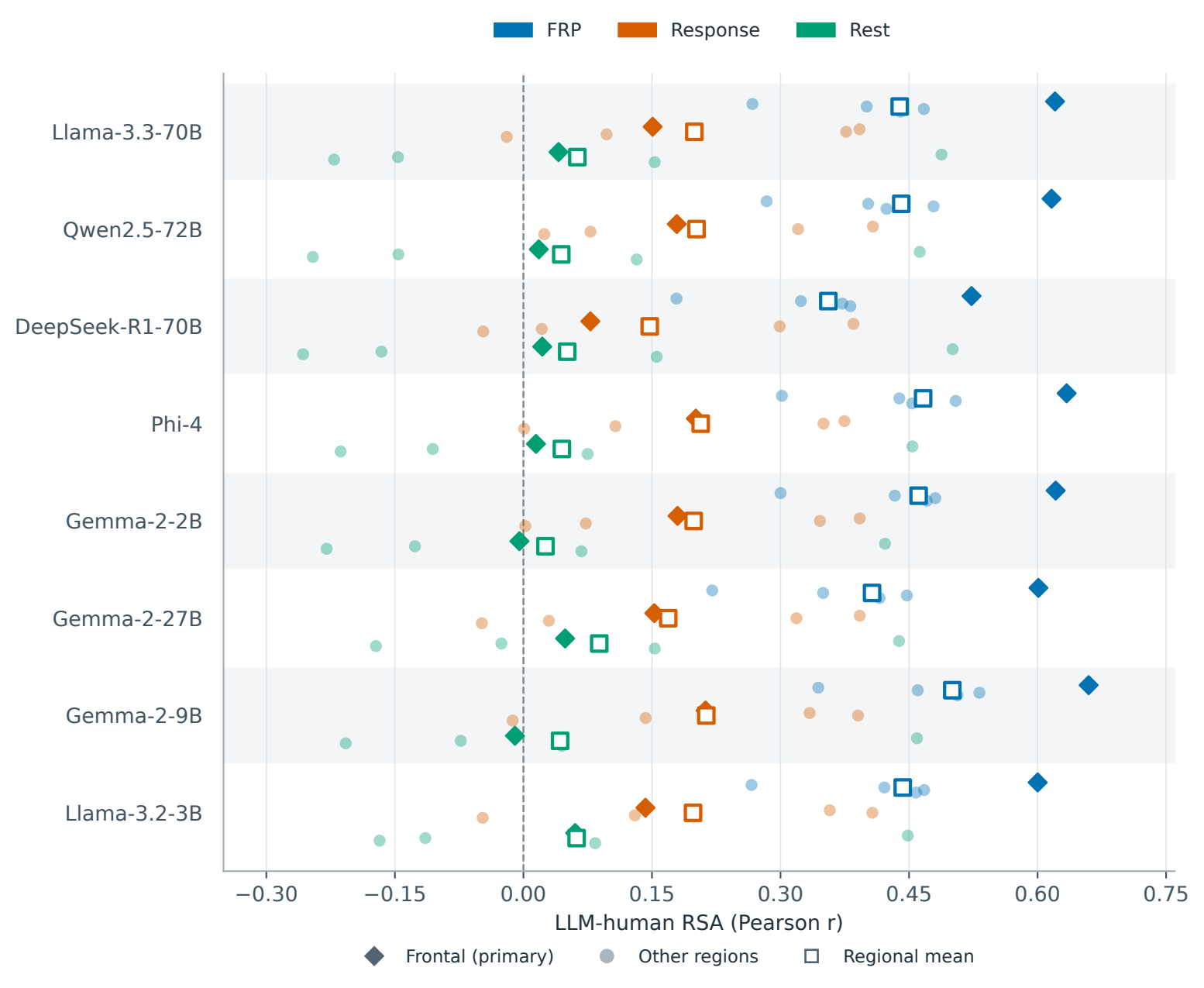

A Task-design reference
AAABAAAB
ABABCDCD
ABBAABBA
ABBACDDC
ABBCABBC
ABCAABCA
ABCDDCBA
ABCDEEDC
Same pattern
Different pattern
B Mean selected-layer RDM
Correlation distance (1 - r)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
C Each model
Mean distance
0.4
0.3
0.2
0.1
0.0
Within pattern
Between patterns

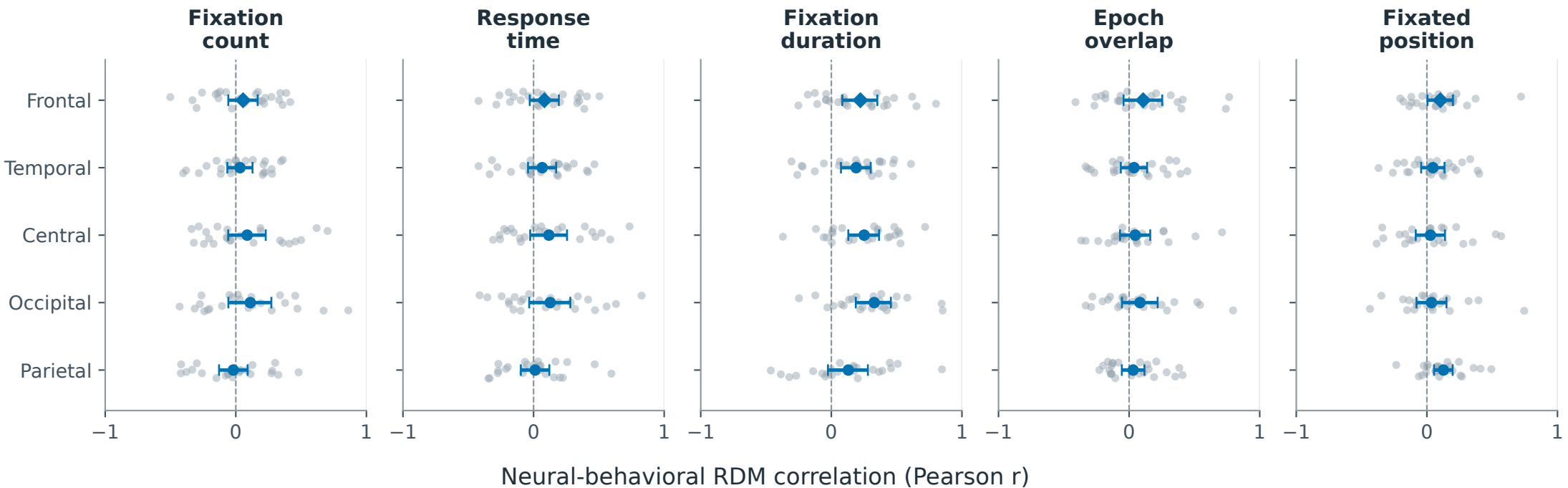

Phi-4
Gemma-2-2B
Gemma-2-9B
Gemma-2-27B
Llama-3.2-3B
Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Peak: 46% depth
Peak: 52% depth
Peak: 49% depth
Peak: 51% depth
Peak: 41% depth
Peak: 39% depth
Peak: 70% depth
Peak: 23% depth
Pearson r with task-design RDM
Relative layer depth
0.0
0.2
0.4
0.6
0.8
0.5
1.0


Frontal
Temporal
Central
Occipital
Parietal
Regional mean
0.61
0.16
0.02
0.27
0.38
0.46
0.40
0.35
-0.11
0.47
0.08
-0.21
0.44
-0.02
0.11
0.44
0.19
0.05
FRP
Response
Rest
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Mean LLM-human RSA (r)


FRP
Response
Rest
Mean LLM-human RSA (Pearson r)
0.6
0.4
0.2
0.0
−0.2
0.61
0.16
0.02
0.27
0.38
0.46
0.40
0.35
-0.11
0.47
0.08
-0.21
0.44
-0.02
0.11
0.44
0.19
0.05
Frontal
Temporal
Central
Occipital
Parietal
Regional mean


FRP
Response
Rest
Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Phi-4
Gemma-2-2B
Gemma-2-27B
Gemma-2-9B
Llama-3.2-3B
−0.3
0.0
0.3
0.6
RSA (Pearson r)
Frontal (primary)
Other regional values
Mean of five regions


FRP
Response
Rest
Llama-3.3-70B
Qwen2.5-72B
DeepSeek-R1-70B
Phi-4
Gemma-2-2B
Gemma-2-27B
Gemma-2-9B
Llama-3.2-3B
−0.30
−0.15
0.00
0.15
0.30
0.45
0.60
0.75
LLM-human RSA (Pearson r)
Frontal (primary)
Other regions
Regional mean


Fixation count
Response time
Fixation duration
Epoch overlap
Fixated position
Frontal
Temporal
Central
Occipital
Parietal
−1
0
1
Neural-behavioral RDM correlation (Pearson r)